

A drone that can fly without wings



Despite small size, this UAV without any exposed rotor blades exhibits powerful monitoring features for safe flight in confined environments (Credit: www.electronicsforu.com)

The Dronut X1 is a bi-rotor ducted drone produced by Cleo Robotics, a Boston based robotics company. Unlike conventional drones, it can navigate through the air with no exposed propellers, making it capable of flying in constricted areas without any worry about its wings colliding with the surroundings. Powered by Qualcomm Snapdragon, the thrust vector technology of the Wi-Fi operated drone offers Dronut X1 high manoeuvrability. It can be controlled with an Android phone by installing its smartphone app. Dronut X1 comes equipped with advanced algorithms, and thanks to its sophisticated sensors, 4k cameras, and LED headlights the drone can collect high-definition lidar data at 30fps in challenging confined spaces having low to negligible light.

Plastic scanner that simplifies plastic recycling

To accelerate the process of plastic sorting and introduce good practices for waste management that can be complied with by businesses and the general public, a fully open source, plastic segregating device has been developed by Jerry de Vos, an industrial design engineer from the Delft University of Technology in the Netherlands. The invention has won the James Dyson Award 2021 for sustainability. The device works by leveraging near-infrared spectrometry (ranging from 750nm to 2500nm) to identify different types of plastics since different types of plastics have different absorbance peaks. The wavelength measurement is done by an integrated InGaAs sensor.

The open source hardware helps to quickly identify different plastic materials and polymer types in recycling facilities (Credit: www.electronicsforu.com)

